

# AHMED EZ-ZOUINE

## Software Engineer · AI & Mobile · Systems & Security

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### SUMMARY

Final-year software engineering student with hands-on experience in AI, Mobile, and Security. Simultaneously pursuing a Master's in Systems Security at Mohammed V University (UM5R) and a professional certification at 42 Network (1337), combining academic depth with intensive project-based engineering. Built and deployed real-world applications serving 900+ users across iOS and Android.

### EDUCATION

#### Master of Science in Computer Engineering and Systems Security

Mohammed V University -(UM5R) - FSR

2024 – 2026

Rabat, Morocco

- **Coursework:** ML, AI, Distributed Systems, Application Security, Cloud Computing and Virtualization, Network Security, Cryptography, Internet of Things (IoT), Data Security and Blockchain Technology, Advanced Web & Mobile Development

#### Digital Technology Architecture

42 School Network – (UM6P), 1337

2023 – 2026

Ben Guerir, Morocco

- **Core Competencies:** Advanced Data Structures & Algorithms, Database Design & Optimization, Software Engineering Principles, Computer Networks & System Architecture, Object-Oriented Programming
- **Projects:** FT\_IRC (IRC server Chat in C++ with sockets) PUSH\_SWAP (Sorting algorithm optimization time complexity) | PHILOSOPHERS (Multithreading & synchronization problem) | MINISHELL (POSIX shell implementation) | CUB3D (Raycasting 3D engine) | INCEPTION (Docker infrastructure)

### PROFESSIONAL EXPERIENCE

#### Software Engineer Intern – AI & Mobile

Uxload

May 2023 – June 2023

Rabat, Morocco

- Built **Diaggy.ai**, a cross-platform mobile health application (React Native, TypeScript, OpenAI API) delivering AI-powered pre-diagnosis to **900+ active users** deployed on **Apple App Store & Google Play Store**
- Collaborated with a cross-functional team of 7 engineers; led code reviews and implemented unit test suites

#### Front-End Developer Intern – UI/UX

StarArt

January 2023 – April 2023

Meknes, Morocco

- Redesigned company website (React.js, Tailwind CSS, Figma), boosting user engagement by **40%** and cutting bounce rate by **25%** through improved UX patterns
- Implemented responsive design and accessibility standards across desktop, tablet, and mobile devices

### TECHNICAL SKILLS

|                       |                                                                       |
|-----------------------|-----------------------------------------------------------------------|
| Languages             | Java, Python, C, C++, JavaScript/TypeScript, SQL                      |
| Web & Mobile          | React, React Native, Flutter, Next.js, Node.js, Express.js, REST APIs |
| Cloud & DevOps        | Docker, Docker Compose, Git, Linux/UNIX                               |
| Databases             | MySQL, MongoDB, Redis, SQL Server                                     |
| Blockchain & Security | Solidity, Smart Contracts, Cryptography (SHA-256, ECDSA), OWASP       |

### PROJECTS

#### Transcendence — Real-Time Multiplayer Game

JavaScript, TypeScript, Figma, Network Programming

- Developed a 3D (three.js) multiplayer Pong game with real-time WebSocket gameplay, input sync, and scoring system
- Designed tournament management system with bracket logic, admin controls, and player ranking
- Containerized full-stack deployment using Docker Compose in collaboration with a team of 3 engineers optimizes performance and ensures a smooth user experience.

#### Smart Agriculture Monitoring System

IOS/Android, RESTful API, JAX-WS, Kafka, IoT Sensors

- Designed a distributed system for real-time monitoring of environmental conditions (temperature, soil moisture, pH) in farms
- Developed data collection, aggregation, and analysis pipelines from multiple IoT sensors
- Built an interactive dashboard with visualization, alerts, and intelligent recommendations for farmers

#### Mobile Anti-Phishing Application

React Native, Expo, TypeScript, SQLite, QR code, Node.js

- Developed a cross-platform (iOS/Android) mobile application that intercepts URLs and QR codes to detect phishing threats before access
- Implemented a local heuristics-based risk scoring engine with regex analysis, typosquatting detection
- Integrated secure WebView preview, TLS validation checks, encrypted local history storage, and privacy-preserving reputation API verification

### LEADERSHIP & VOLUNTEER EXPERIENCE

#### Statistical Data Collector – National Census

RGPH 2024 – Morocco National Population Census

2024 (45 days)

Morocco

- Contributed to Morocco's national population census project, collecting and validating demographic data
- Developed strong communication and cross-cultural collaboration skills working with diverse communities

### LANGUAGES & CERTIFICATIONS

**Certifications:** Google Cybersecurity Professional Certificate (6 months) | Google Data Analytics Pro Certificate (6 months—Expected 2026)

**Languages:** Arabic (Native) | French (Intermediate) | English (Professional Working Proficiency)